

NPM Commands

A FULL REFERENCE

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``npm`` stands for Node Package Manager. It is the default package manager for Node.js.

It is used for installing packages, managing dependencies, running project scripts, and publishing packages to the ``npm`` registry.



INIT & PROJECT SETUP

Command	Description
<code>npm init</code>	Create a new package.json file interactively.
<code>npm init -y</code>	Create package.json with defaults.
<code>npx <package> <name></code>	Run a CLI scaffold without installing it globally.
<code>npm install -g <package></code>	Install a package globally (useful for CLI tools).
<code>npm uninstall -g <package></code>	Remove a globally installed package.
<code>npm link</code>	Symlink a local package into global node_modules for local development.
<code>npm unlink</code>	Remove a previously created link.
<code>npm bin</code>	Print path to local node_modules/.bin.



INSTALL & REMOVE PACKAGES

Command	Description
<code>npm install <package></code>	Install a package as a dependency.
<code>npm i <package></code>	Short for npm install.
<code>npm install --save-dev <package></code>	Install a package as a development dependency.
<code>npm i -D <package></code>	Short for npm install --save-dev.
<code>npm uninstall <package></code>	Remove a package and update package.json.
<code>npm prune</code>	Remove extraneous packages not listed in package.json.
<code>npm dedupe</code>	Remove duplicate dependencies.
<code>npm outdated</code>	Show which dependencies are newer than the installed versions.
<code>npm ls --depth=0</code>	List top-level installed packages for the current project.

Tip: Use `rm -rf node_modules package-lock.json && npm install` to recreate `node_modules` and lockfile to recover from dependency issues.

RUNNING SCRIPTS

Command	Description
<code>npm run build</code>	Run the build script defined in package.json.
<code>npm run test -- --watch</code>	Run test with additional args passed after --.
<code>npm start</code>	Shortcut to run the start script (if defined).
<code>npm test</code>	Shortcut to run the test script (if defined).
<code>npx eslint src --fix</code>	Run a package binary from node_modules/.bin without installing it globally.
<code>npx --yes <package></code>	Run a package non-interactively and accept prompts automatically.
<code>npm explore <package> -- <cmd></code>	Run a command inside a dependency's package folder.



CI / LOCKFILE COMMANDS

Command	Description
npm ci	Install exactly from package-lock.json for reproducible CI installs.
npm install	Install dependencies and generate/update package-lock.json.

VERSIONING & PUBLISHING

Command	Description
<code>npm version patch</code>	Bump patch version, update package.json, and create a git tag.
<code>npm pack</code>	Create a .tgz tarball of the package for local testing.
<code>npm publish --dry-run</code>	Preview what would be published without uploading.
<code>npm publish --access public</code>	Publish the package to the npm registry (use --access for scoped packages).
<code>npm dist-tag</code>	Manage distribution tags (e.g., npm dist-tag add <package>@1.2.3 latest).
<code>npm view <package> version</code>	Show metadata for a package on the npm registry.

AUDIT & TROUBLESHOOTING

Command	Description
<code>npm audit</code>	Run a security audit of installed dependencies.
<code>npm audit fix</code>	Attempt to automatically fix vulnerabilities.
<code>npm cache clean --force</code>	Clear npm cache when troubleshooting cache-related issues.
<code>npm rebuild</code>	Rebuild native modules after Node or dependency changes.
<code>npm prune</code>	Remove extraneous packages not listed in package.json.
<code>npm dedupe</code>	De-duplicate the dependency tree to reduce duplicate installs.
<code>npm doctor</code>	Run npm health checks and configuration diagnostics.

AUTH & CONFIG

Command	Description
<code>npm login</code>	Authenticate to the npm registry (prompts for credentials).
<code>npm logout</code>	Remove locally stored npm credentials.
<code>npm whoami</code>	Show the currently logged-in npm username.
<code>npm token create</code>	Create an automation token for CI (handle securely).
<code>npm token revoke</code>	Revoke an existing npm token.
<code>npm set registry https://registry.npmjs.org/</code>	Point npm to a specific registry URL.
<code>npm config set init-author-name "Your Name"</code>	Set npm configuration values (persisted in user config).